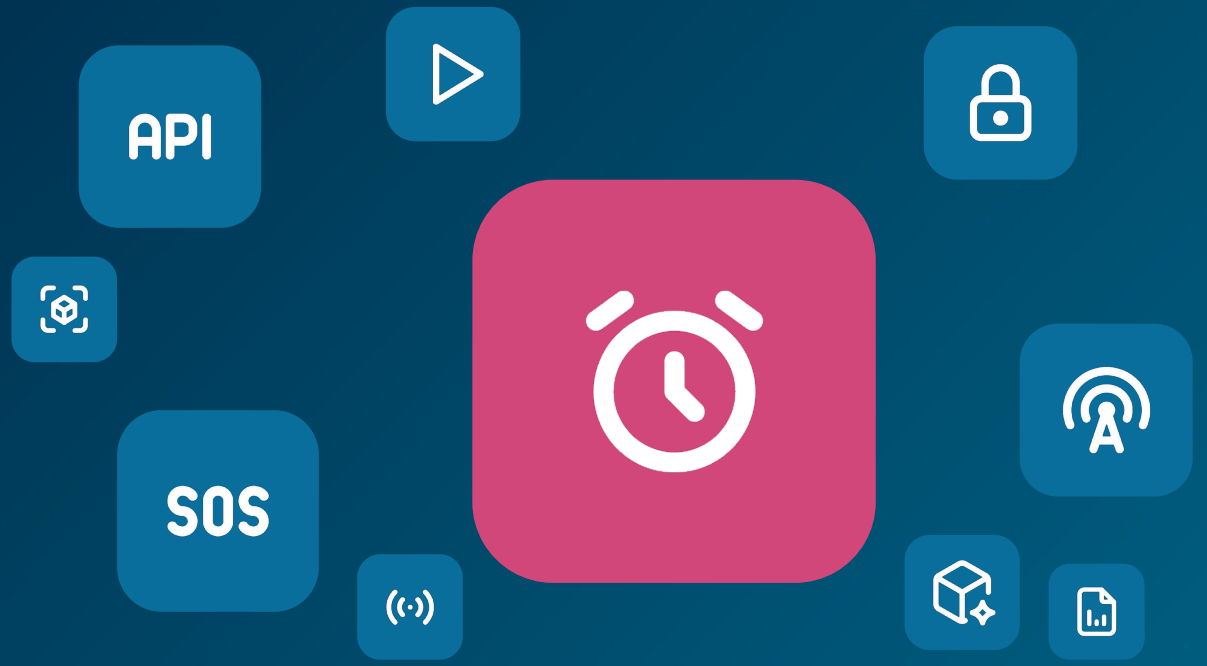
5G MAG The Media Connectivity Association



OVERVIEW AND UPDATES · CONNECTED MEDIA PRODUCTION

Time-Sensitive Communications (TSC)

Deterministic, low-jitter transport for tightly synchronised professional media

3GPP

NOVEMBER 2026 – www.5g-mag.com

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About the project

The 5G system as a bridge between wired Time-Sensitive Networking segments

TS 23.501

3GPP system architecture

The 5G system modelled as a TSN bridge, NW-TT and DS-TT, and time synchronisation.

TS 23.502

3GPP procedures

The procedures of Time-Sensitive Communications.

TS 29.565

3GPP stage 3

The services of the Time Sensitive Communication and Time Synchronization Function (TSCTSF). IEEE 802.1 TSN standards are referred to throughout.

FIND OUT MORE **Technology** **Specifications**

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Control plane

How 5G carries deterministic traffic and shared timing between wired TSN segments

AT THE UPF

NW-TT

The Network-side TSN Translator at the UPF. It terminates the wired TSN network and holds the bridge management information.

IN THE 5G CORE

TSCTSF

The network function through which deterministic QoS and time synchronisation are requested. An AF outside the trust domain reaches it through the NEF.

IN THE TSN NETWORK

TSN CNC

Computes schedules across the 5G bridge with ordinary TSN tooling; the 5G system reports its bridge capabilities to it.

The 5G system is modelled as one or more virtual TSN bridges, so wired TSN tooling can schedule across the wireless hop.

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Specifications in scope

The standards that 5G TSC interworks with and the media workflow it carries

3GPP service requirements and system architecture

- TS 22.104
- TR 22.804
- TS 23.501
- TS 23.502
- TR 23.700-25
- TS 29.565
- TS 29.522

IEEE TSN Time-Sensitive Networking

- 802.1AS timing and synchronisation
- 802.1Q bridges
- 802.1BA Audio Video Bridging
- 802.1CB frame replication

SMPTE IP media production

- ST 2110-10 timing
- ST 2110-20 video
- ST 2110-21 traffic shaping
- ST 2110-30 audio
- ST 2110-40 ancillary data

TSC in a media context usually runs over a private 5G deployment, so it builds on Non-Public Networks.

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How it grew

What each 3GPP release added to Time-Sensitive Communications

Release 16

The 5G system as a TSN bridge, with DS-TT and NW-TT. gPTP is forwarded with residence-time correction; NR gains reference timing delivery and TSC assistance information.

Release 17

The TSCTSF and generalised time synchronisation (several IEEE 802.1AS and IEEE 1588 clock roles), exposure to AFs through the NEF, survival time, propagation delay compensation.

Release 18

Reporting of the timing synchronisation status, interworking with a TSN transport network, scheduling adaptation from RAN feedback, TSN charging, DetNet extensions.

Release 19

QoS monitoring capability enhancement (TEI19_QME, TS 29.565); time synchronisation in the MBS architecture of 5G Media Streaming.

Release 20

Release 20 (work plan of 28 September 2026): change requests to TS 29.565 only. Unverified: no TR 21.920 yet.

WHERE WE STAND

Time-Sensitive Communications (TSC)

BROUGHT TO STUDY | ON-GOING | OUTCOMES

- **Report: Time synchronization services for media production over 5G networks**
- **Report: Time synchronization and media streams for content production using SMPTE ST 2110 standards**

RELATION WITH SDOS AND INDUSTRY FORA **3GPP**

The Media Connectivity Association

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